HTML : Hyper Text Markup Language
.html extension is used to save a file

Structure:

```
<!DOCTYPE html>
<html>
   <head>
      Meta  Title  Link
   </head>
   <body>
      <div>
         Img h1  p  div
      </div>
   </body>
</html>
```

Code1:

```
<html>
   <head>
      <title>My Website</title>
   </head>
   <body>
      <h1>Welcome to HTML</h1>
   </body>
</html>
```

HTML Elements: Building Blocks of Webpages

<start> content </end>

**
** single line break
<hr> To divide or separate Documents(a line)
These Both are Empty Elements.

HTML Headings:

<h1> </h1> and so on most important

```html
<!DOCTYPE html>

<html>

<head>
  <title>My Website</title>
</head>

<body>
  <h1>H1: Heading</h1>
  <h2>H2: Heading</h2>
  <h3>H3: Heading</h3>
  <h4>H4: Heading</h4>
  <h5>H5: Heading</h5>
  <h6>H6: Heading</h6>
</body>

</html>
```

H1: Heading

H2: Heading

H3: Heading

H4: Heading

H5: Heading

H6: Heading

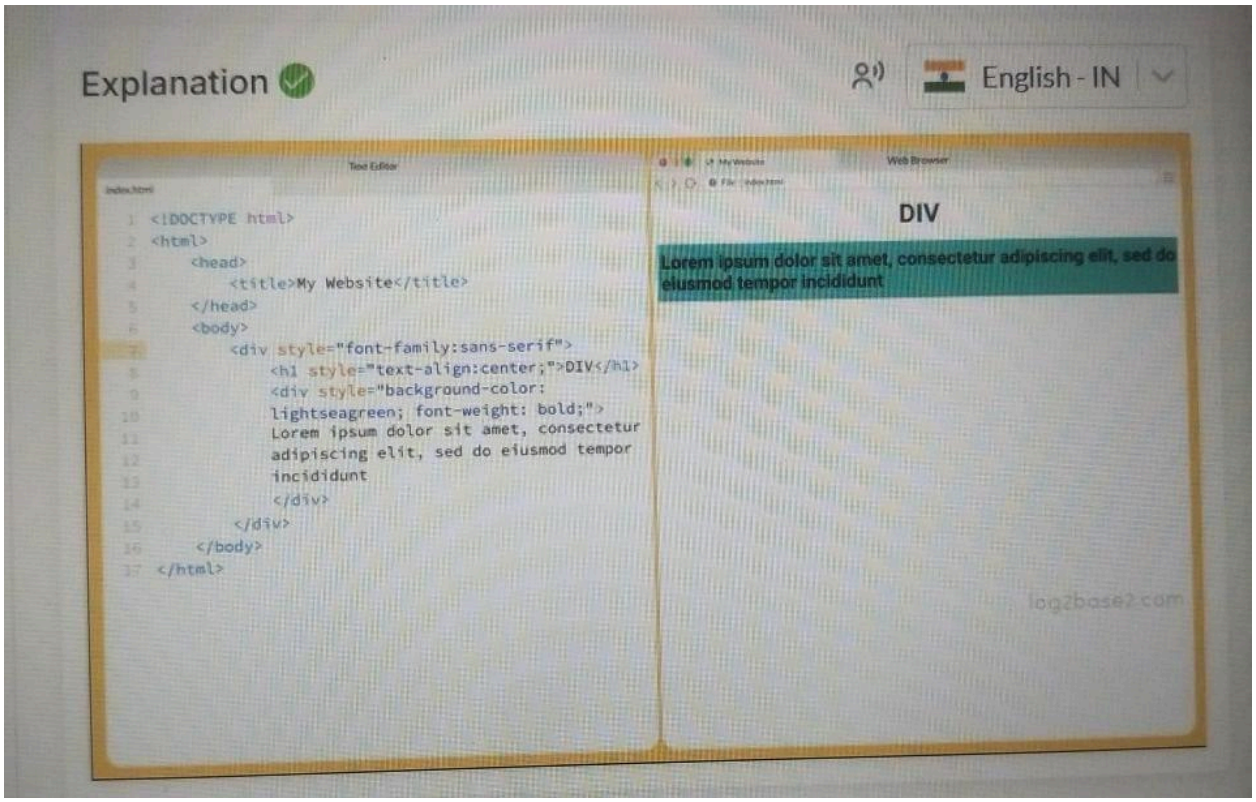
HTML Paragraphs : content

```html
<!DOCTYPE html>

<html>

<head>
  <title>My Website</title>
</head>

<body>
  <p>This is a paragraph. It
contains several lines of text that
will be displayed together as a
block.</p>
 <p>You can add as many paragraphs
as you like to your HTML page by
using the "p" tag.</p>
</body>

</html>
```

This is a paragraph. It contains several lines of text that will be displayed together as a block.

You can add as many paragraphs as you like to your HTML page by using the "p" tag.

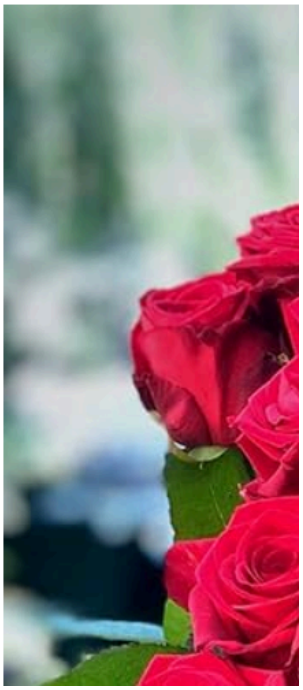
Div Tag: Used to Create Sections or Divisons .



IMG TAG: Used to insert Images

Src : To Specify Location of an image

```
<!DOCTYPE html>

<html>

<head>
  <title>My Website</title>
</head>

<body>
  <img src="https://m.media-amazon.com/images/I/81OXEQrFPTL._AC_UF1000
</body>

</html>
```



Absolute and Relative Path: These are the two ways to include img src.
Relative paths are used to specify the location of the image file relative to the location of the HTML file, while absolute paths are used to load files from external domains.
1)complete path
Protocol Domain_name File_path (https:// example.com /images/image.jpg**)**

Our Domain - - - - - - - - - - - Different Domain
Exact Path

2)Load images from our own Domain , not from external Domains

1.**Same Directory** : if img is stored in the same directory of HTML file.
2.**sub Directory**: if img is stored in sub directory relative to HTML file
We need to specify sub folder name and image name in src of img tag

Webpage -> index.html **<img src="images/icon.png">**
-> images ->icon.png

3.**Parent Directory**: we need to use .../imagename **<img src="../icon.png">**
For multiple , <img src="../../icon.png">

Webpage(Parent Folder) ->icon.png
| -> **HTML(**Sub Folder**) ->** index.html
|----------|

Importance of Alt Text in Images:
The browser displays the "alt" text 'Icon Image' along with a broken image icon
If URL of an image is wrong or image is not displayed then a blocked image is displayed . so ALT is used as an Alternative text for the image.

HTML Attributes & Image Height/ Width Attributes:
<start > attribute <end>
Provides **Additional Info**
Specified in **Start tag**
Come in name / value pairs like **name ="value"**
Can enhance the **functionality and appearance** of HTML elements

```html
<!DOCTYPE html>

<html>

<head>
  <title>My Website</title>
</head>

<body>
  <img src="https://m.media-amazon.com/images/I/81OXEQrFPTL._AC_UF1000
  height="500px" width="400px" alt="Rose ">
</body>

</html>
```

HTML Links:

Anchor Tag: Used to create links in the webpage

Navigates to another websites and all

<a href=" url"> GOOGLE </a>

```html
<!DOCTYPE html>

<html>

<head>
  <title>My Website</title>
</head>

<body>
  <a
href="https://www.wikipedia.org">Wik
ipedia</a>
</body>

</html>
```

LIVE

Wikipedia

HTML Lists: To group items together

Ordered List: Number <ol><li> </li></ol>

Unordered List: Bullet points <ul> <li> </li> </ul>

Where , li means List Item

```html
<html>

<head>
  <title>My Website</title>
</head>

<body>
  <h1>HTML Lists</h1>
  <hr>
  <h2>Sample Unordered
List</h2>
  <ul>
    <li>Item 1</li>
    <li>Item 2</li>
    <li>Item 3</li>
  </ul>
  <hr>
  <h2>Sample Ordered
List</h2>
  <ol>
    <li>First item</li>
    <li>Second item</li>
    <li>Third item</li>
  </ol>

</body>

</html>
```

HTML Lists

Sample Unordered List

- Item 1
- Item 2
- Item 3

Sample Ordered List

1. First item
2. Second item
3. Third item

HTML Tables:

<table > </table>

<th>Table Header

<tr> Table row

<td> Table Data

Table Border : Border ="1"

```html
<!DOCTYPE html>

<html>

<head>
  <title>My Website</title>
</head>

<body>
  <h2>Sample HTML Table</h2>

  <table border="1">
    <tr>
      <th>Name</th>
      <th>Age</th>
      <th>Country</th>
    </tr>
    <tr>
      <td>John</td>
      <td>21</td>
      <td>United States</td>
    </tr>
    <tr>
      <td>Ben</td>
      <td>19</td>
      <td>Germany</td>
    </tr>
    <tr>
```

Sample HTML Table

Name	Age	Country
John	21	United States
Ben	19	Germany
Nick	20	India

HTML Forms:

"for" is used to bind the label and input elements in an HTML form

```html
<!DOCTYPE html>

<html>

<head>
  <title>My Website</title>
</head>

<body>
  <h2>Sample HTML Form</h2>

  <form>
    <label for="email">Email:
</label><br>
    <input type="text" id="email"
name="email"><br>
    <label for="password">Password:
</label><br>
    <input type="password"
id="password" name="password"><br>
<br>
    <input type="submit"
value="Sign in">
  </form>

</body>

</html>
```

Sample HTML Form

Email:

Password:

Sign in

Registration form 👍

```
<!DOCTYPE html>
<html>

<head>
  <title>
    Job Registration Page
  </title>
</head>

<body>
  <h1 style="text-align: center;"> Job Registration form</h1>
  <form action="/action_page.php">
    <label> Firstname </label>
    <input type="text" name="firstname"> <br> <br>
    <label> Lastname: </label>
    <input type="text" name="lastname"> <br> <br>
    <label for="birthday">Birthdate:</label>
    <input type="date" id="birthdate" name="birthdate">
    <br>
    <br>
```

```
    <label>
      Course :
    </label>
    <select>
      <option value="Course">Course</option>
      <option value="BCA">BCA</option>
      <option value="BBA">BBA</option>
      <option value="B.Tech">B.Tech</option>
      <option value="MBA">MBA</option>
      <option value="MCA">MCA</option>
      <option value="M.Tech">M.Tech</option>
    </select>
    <br>
    <br>
    <label>
      Gender :
    </label><br>
    <input type="radio" name="Gender" value="male" /> Male <br>
    <input type="radio" name="Gender" value="female" /> Female <br>
    <input type="radio" name="Gender" value="other" /> Other
    <br>
    <br>
    <label>
      Preferred Locations:
    </label>
    <br>
    <input type="checkbox" id="India">
    <label for="India"> India</label><br>
    <input type="checkbox" id="UK">
    <label for="UK"> UK</label><br>
    <input type="checkbox" id="USA">
    <label for="USA"> USA</label><br>
    <input type="checkbox" id="Germany">
    <label for="Germany"> Germany</label><br>
    <input type="checkbox" id="France">
    <label for="France"> France</label><br><br>
    <label>
      Phone :
    </label>
    <input type="text" name="country code" value="+91" size="2">
    <input type="number" name="phone" maxlength="10" size="10">
    <br> <br>
    Address:
    <br>
    <textarea value="address" rows="5" cols="25">
  </textarea>
```

```
  <br> <br>
  <label for="resume">Upload your resume:</label>
  <input type="file" id="resume" name="resume"><br><br>
  <br>
  <label for="linkedin">Add your Linkedin url:</label>
  <input type="url" id="linkedin" name="linkedin">
  <br><br>
  <label for="vol">Rate your English proficiency:</label>
  <input type="range" id="vol" min="0" max="10">
  <br><br>
  <label for="email">
    Email:
  </label>
  <input type="email" id="email" name="email" />
  <br> <br>
  <button type="submit"> Submit</button>
  <button type="reset">Reset</button>
 </form>
</body>
</html>
```

Job Registration form

Firstname

Lastname:

Birthdate: mm / dd / yyyy

Course : Course

Gender :
- Male
- Female
- Other

Preferred Locations:
- India
- UK
- USA
- Germany
- France

Phone : +91

Address:

Upload your resume: Choose File No file chosen

Add your Linkedin url:

Rate your English proficiency:

Email:

Submit Reset

Form Validation:

```
<!DOCTYPE html>
<html>

<head>
  <title>Form Validation</title>
</head>

<body>
  <h1>Registration Form</h1>
  <form>
    <label for="name">Name:</label>
    <input type="text" id="name" name="name" required placeholder="John"><br><br>
    <label for="email">Email:</label>
    <input type="email" id="email" name="email" required placeholder="abc@xyz.com"><br><br>
    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required minlength="8"><br><br>
    <label for="age">Age:</label>
    <input type="number" id="age" name="age" required min="18" max="100"><br><br>
    <label for="phoneNumber">Phone Number:</label>
    <input type="tel" id="phoneNumber" name="phoneNumber" pattern="[0-9]{10}" required
placeholder="e.g., 1234567890"><br><br>
    <label for="birthDate">Birth Date:</label>
    <input type="date" id="birthDate" name="birthDate" required max="2010-01-01"><br><br>
    <label for="website">Website:</label>
    <input type="url" id="website" name="website" required
placeholder="https://www.example.com"><br><br>
    <button type="submit">Register</button>
  </form>
</body>

</html>
```